



STM32F3 Technical Training

For reference only

Refer to the latest documents for details



STM32F30x Specific features/peripherals



Timers' enhancements in STM32F30X

Channel-level enhancements

4

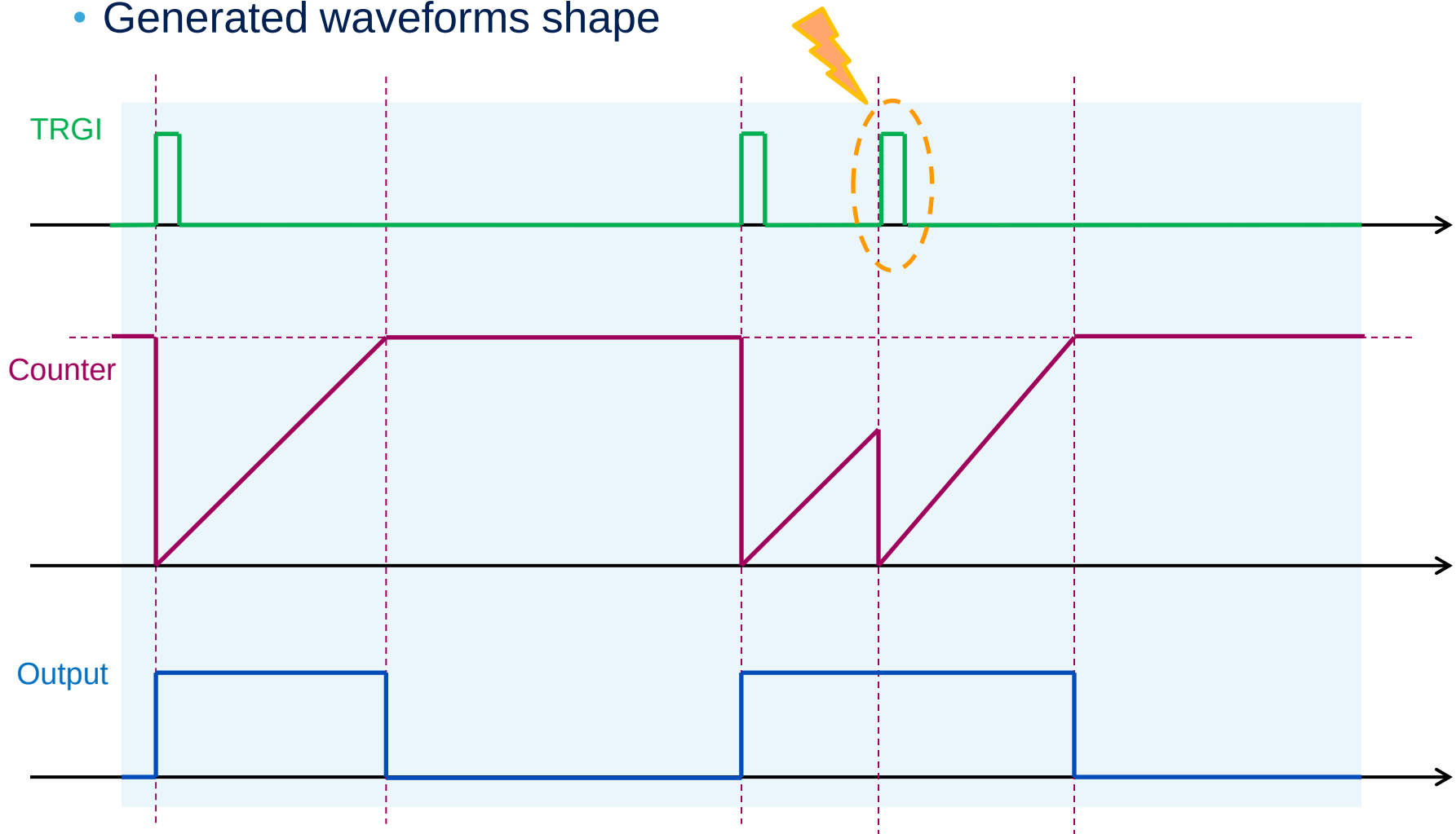
- Up to 6 channels on Advanced control timers:
 - Up to 4 channels with input/output stages (as in TIM2/3/4)
 - Channels remain compatible with those on existing products' timers
 - New features:
 - More complex waveforms generation
 - Enhanced triggering capability
 - More channel's modes
 - Multitude of coupling scenarios between channels
 - 2 extra channels (only on TIM1/8)
 - Internal channels
 - Not wired to GPIOs
 - Used within the Timer itself for complex waveform generation
 - Routed to the ADC triggering logic (via Timer's TRGO output)
 - Compare-and-PWM-modes-only channels
 - No capture modes
 - No DMA channels nor Interrupt request lines
 - High coupling with channels 1,2,3 and4
 - Enhance waveform generation on those channel: More complex waveforms can be generated
 - Enhanced triggering mechanism: ADC oriented triggering mechanism
 - Designed to meet many Motor Control applications' requirements

Retriggerable One Pulse Mode (1/2)

(Not available in TIM16/17)

5

- Generated waveforms shape



Retriggerable One Pulse Mode (2/2)

(Not available on TIM16/17)

6

- Available on Channel 1, 2, 3 and 4
- Different from the existing One Pulse mode:
 - The outputted pulse starts as soon as a trigger active edge is detected
 - The pulse length is extended if a new active edge is detected
- Pulse length is set using the ARR register
 - For Up-counting mode, CCRx register has to be set to zero
 - For Down-counting mode, CCRx register has to be set to ARR value
- Configuration sequence
 - Set the timer to **slave** mode: the **Combined Reset+Trigger** mode shall be used
 - Select the Retriggerable One Pulse mode through the OCxM[3:0] bit field
 - Retriggerable OPM mode 1
 - Retriggerable OPM mode 2

Channels Coupling (1/2)

7

- Two coupling schemes:
 - Adjacent channels coupling:
 - Channel1 and Channel2 coupling
 - Channel3 and channel4 coupling
 - Enhanced channels coupling (feature used by Motor Control applications)
 - Channel5 and Channel1
 - Channel5 and Channel2
 - Channel5 and Channel3
- Flexible coupling mechanism on adjacent channels
 - Channels coupling output can be directed to one channel or to both of them
- Generated Waveforms' shape
 - Frequency control through TIMx_ARR register value
 - Phase-shift (delay) control through one of the two channels' TIMx_CCR register
 - Pulse-length (duty-cycle) control through the second channels' TIMx_CCR register

Channels Coupling (2/2)

8

- Available PWM modes

- Each channel among the first four channels can be configured in one of the following PWM modes
- Asymmetric and Combined PWM modes are applicable on coupled channels only

	PWM mode 1	PWM mode 2
Independent	OCxM[3:0] = 4b'0110'	OCxM[3:0] = 4b'0111'
Asymmetric	OCxM[3:0] = 4b'1110'	OCxM[3:0] = 4b'1111'
Combined	OCxM[3:0] = 4b'1100'	OCxM[3:0] = 4b'1101'

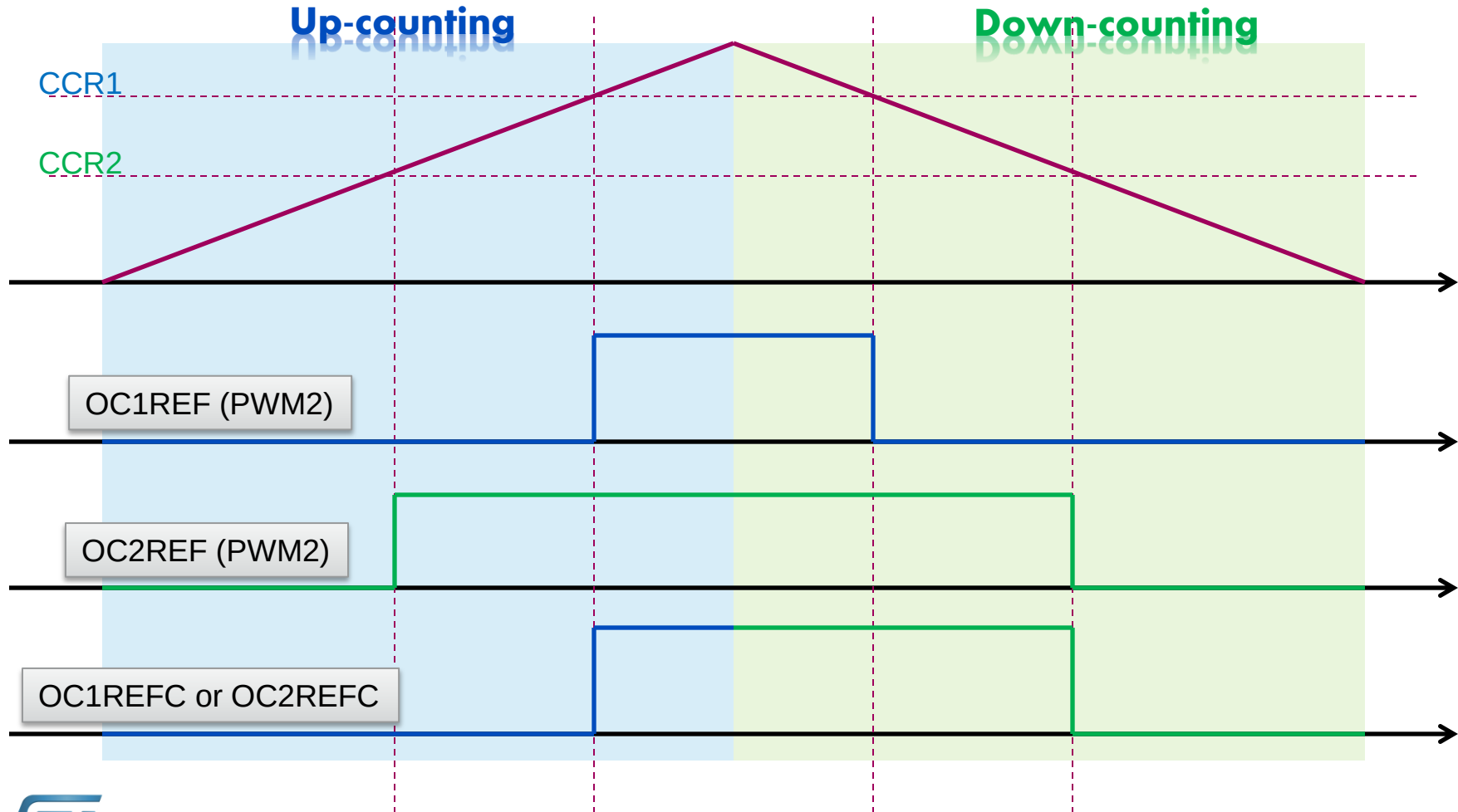
Coupling between channels is activated

Asymmetric PWM mode (1/3)

(Not available in TIM15/16/17)

9

- Output waveform shape

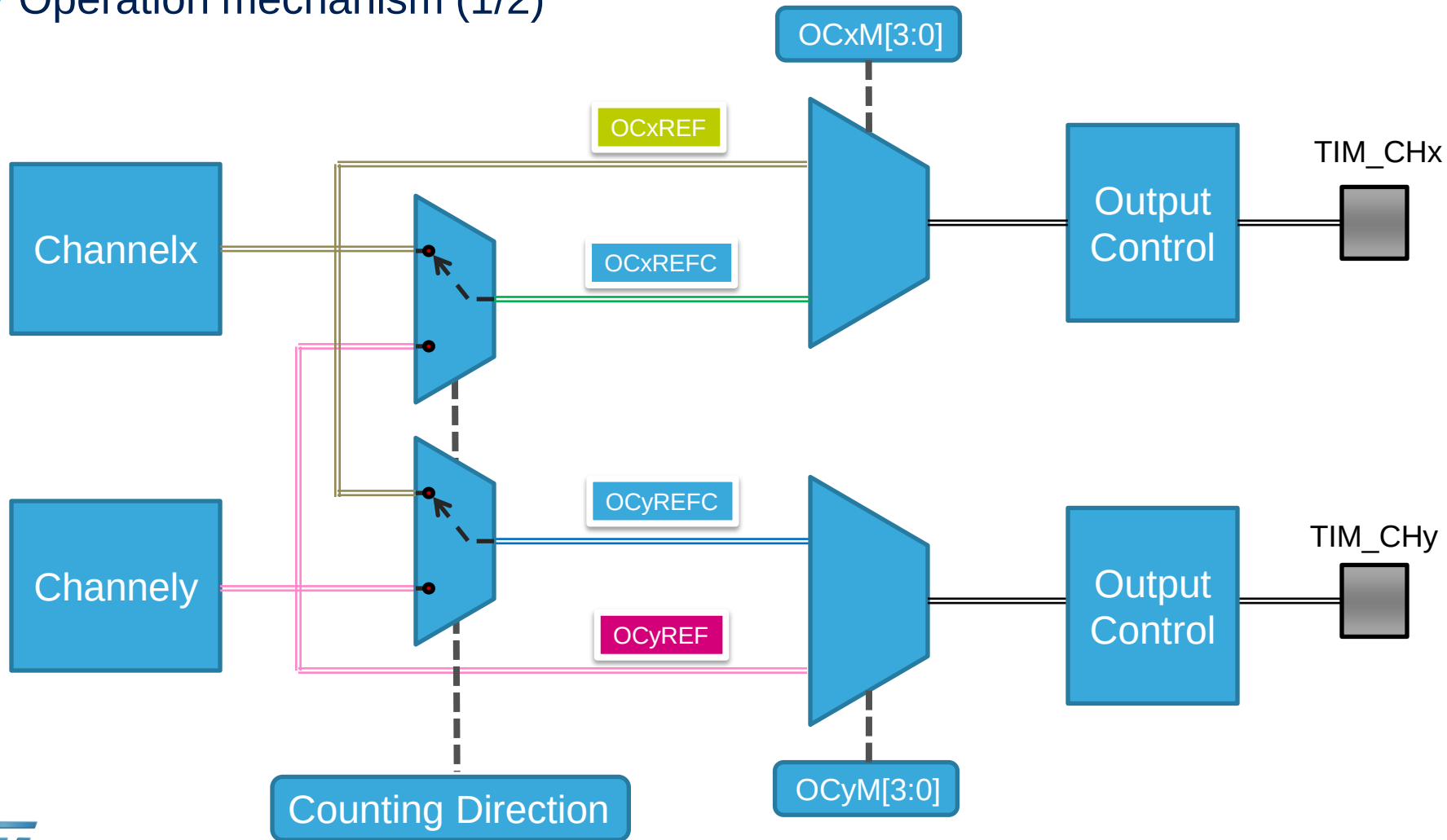


Asymmetric PWM mode (2/3)

(Not available in TIM15/16/17)

10

- Operation mechanism (1/2)



Asymmetric PWM mode (3/3)

(Not available in TIM15/16/17)

11

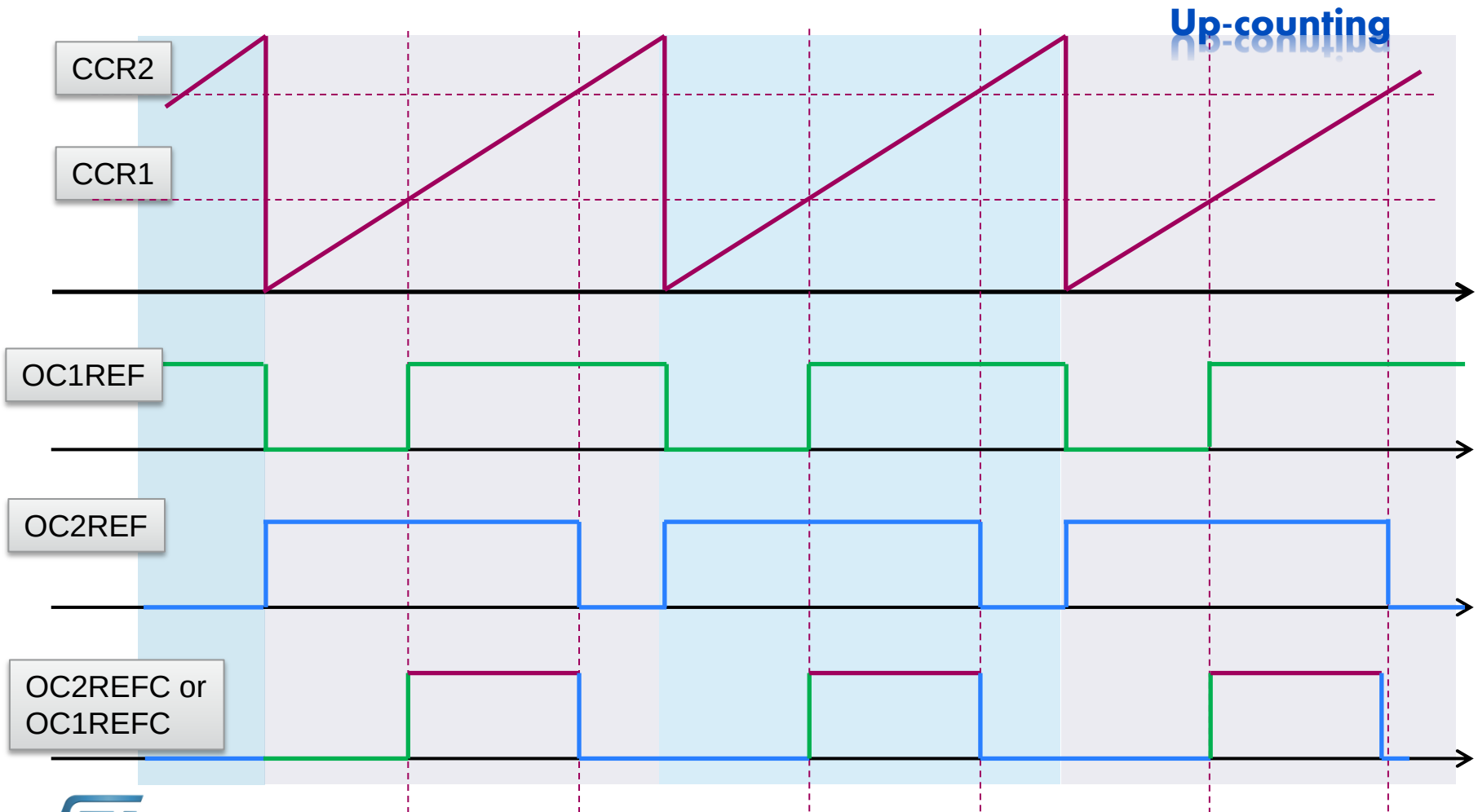
- Operation mechanism (2/2)
 - The counting direction selects which channel output to be directed to OCxREFC
 - Coupled channel has to be configured in the same PWM mode
- Center-aligned counting mode required
 - Asymmetric mode is effective only when the timer is configured to count in center-aligned mode
- Available on the following channel couples:
 - (Channel1, Channel2)
 - (Channel3, Channel4)
- Two Asymmetric PWM mode are available
 - Asymmetric PWM1 mode
 - Asymmetric PWM2 mode

Combined PWM mode (1/5)

(Not available in TIM16/17)

12

- Output waveform shape (Logical And)

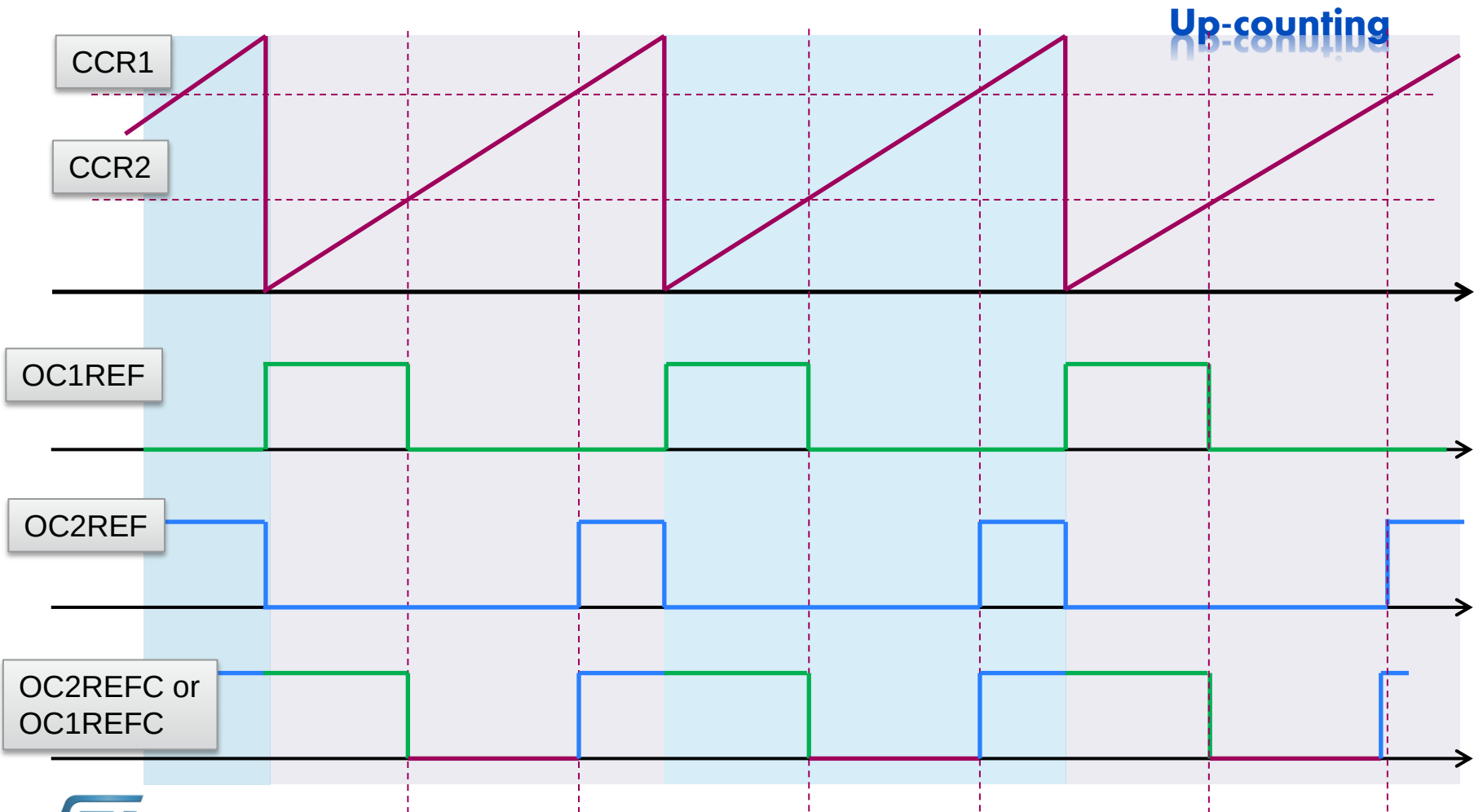


Combined PWM mode (2/5)

(Not available in TIM16/17)

13

- Output waveform shape (Logical Or)

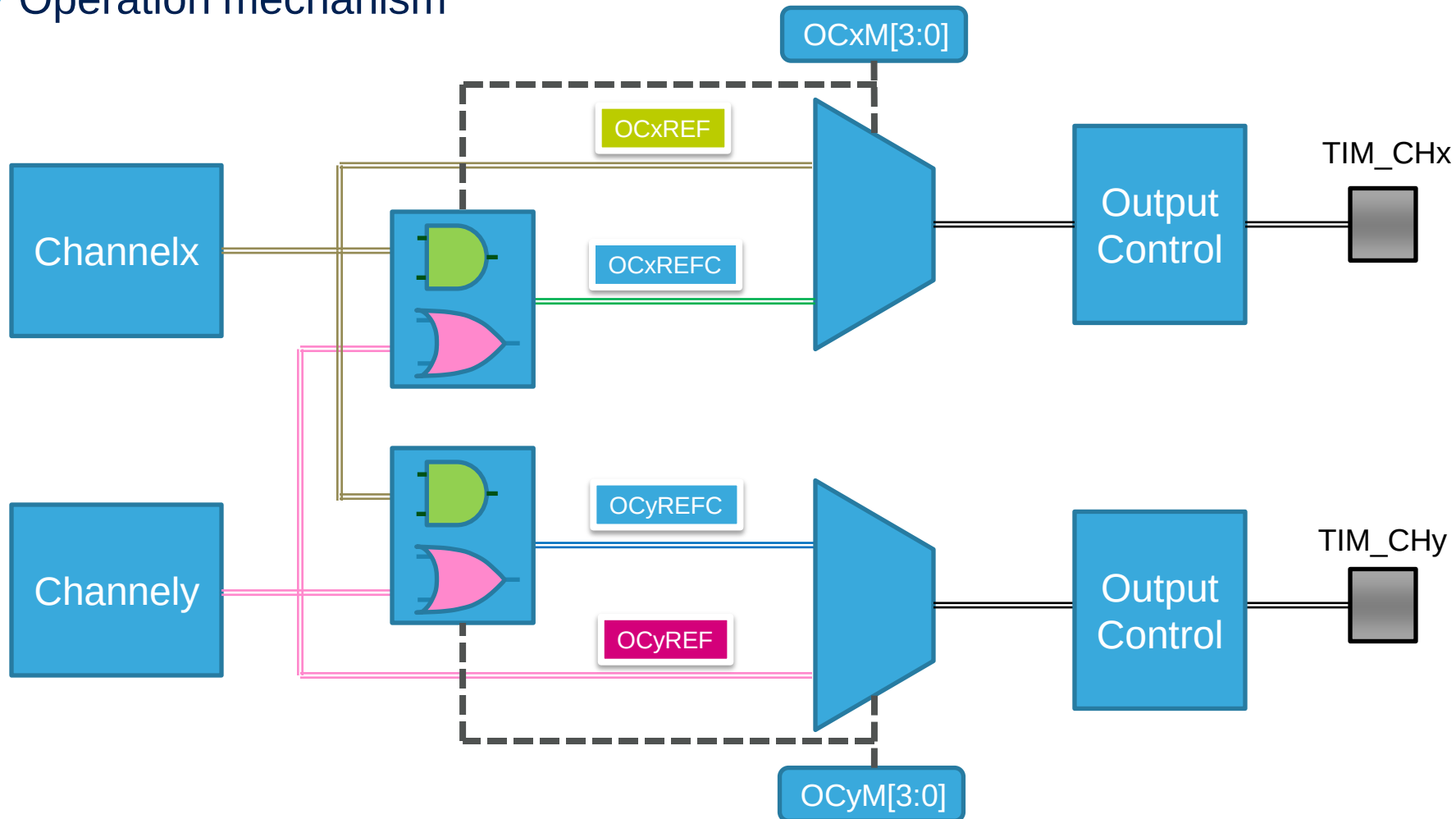


Combined PWM mode (3/5)

(Not available in TIM16/17)

14

- Operation mechanism



Combined PWM mode (4/5)

(Not available in TIM16/17)

15

- Two logical operators coupling modes:
 - Logical And
 - Logical Or
- Two Combined PWM mode are available
 - Combined PWM1 mode
 - Combined PWM2 mode
- Different PWM mode on each channel
 - In order to get the desired output, the two coupled channels has to be configured with different PWM modes: PWM1 and PWM2
 - If the same PWM mode is configured on both channels, the output signal waveform is similar to one of the two channels waveforms depending on the Logical Operator applied

Combined PWM mode (5/5)

(Not available in TIM16/17)

16

- Configuration sequence
 - Configure the two coupled channels on different PWM modes
 - Configure one channel or both coupled channels to output a logical combination of the channels' waveforms
- Counting mode independent:
 - Acts on Edge-aligned counting mode
 - Acts on Center-aligned counting mode
- Available on the following channel couples:
 - (Channel1, Channel2)
 - (Channel3, Channel4)

Channels 5&6 features

(Only TIM1 & TIM8)

17

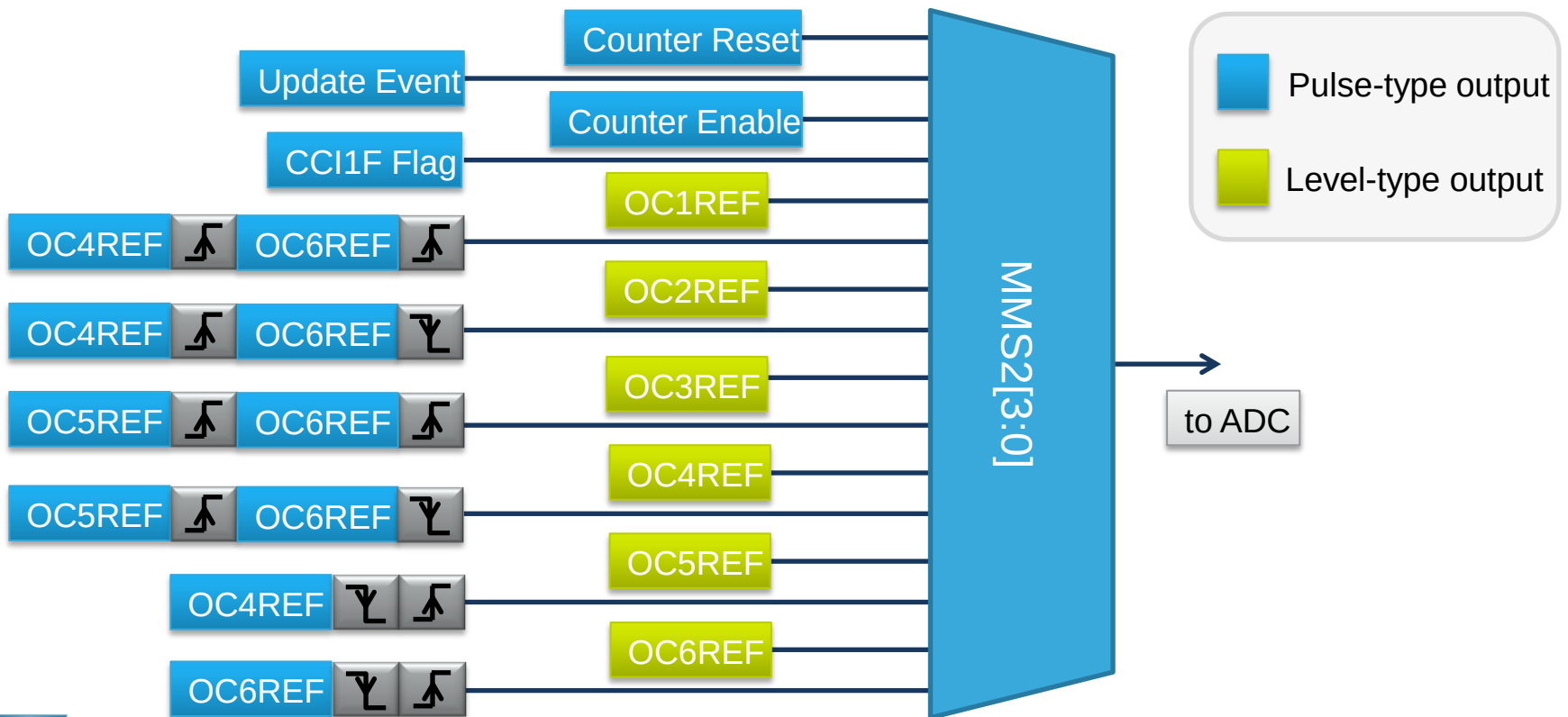
- Channels 5&6 characteristics:
 - Only available on advanced control Timers: TIM1 & TIM8
 - Compare-and-PWM-modes-only channels
 - Internal channels (no external output)
- Channel 5&6 use cases:
 - Can be used to generate more complex waveforms when combined with other channels (applicable for Channel5 only)
 - Can be used to trigger ADC conversion (many triggering scenarios)
- Compatible with the first four channels' implementation
 - Same control registers (for implemented features)
 - Same control bit-fields' structure (for implemented features)
- Typical use case
 - Used by single-shunt current measurement applications

Enhanced Triggering mechanism

(Only TIM1 & TIM8)

18

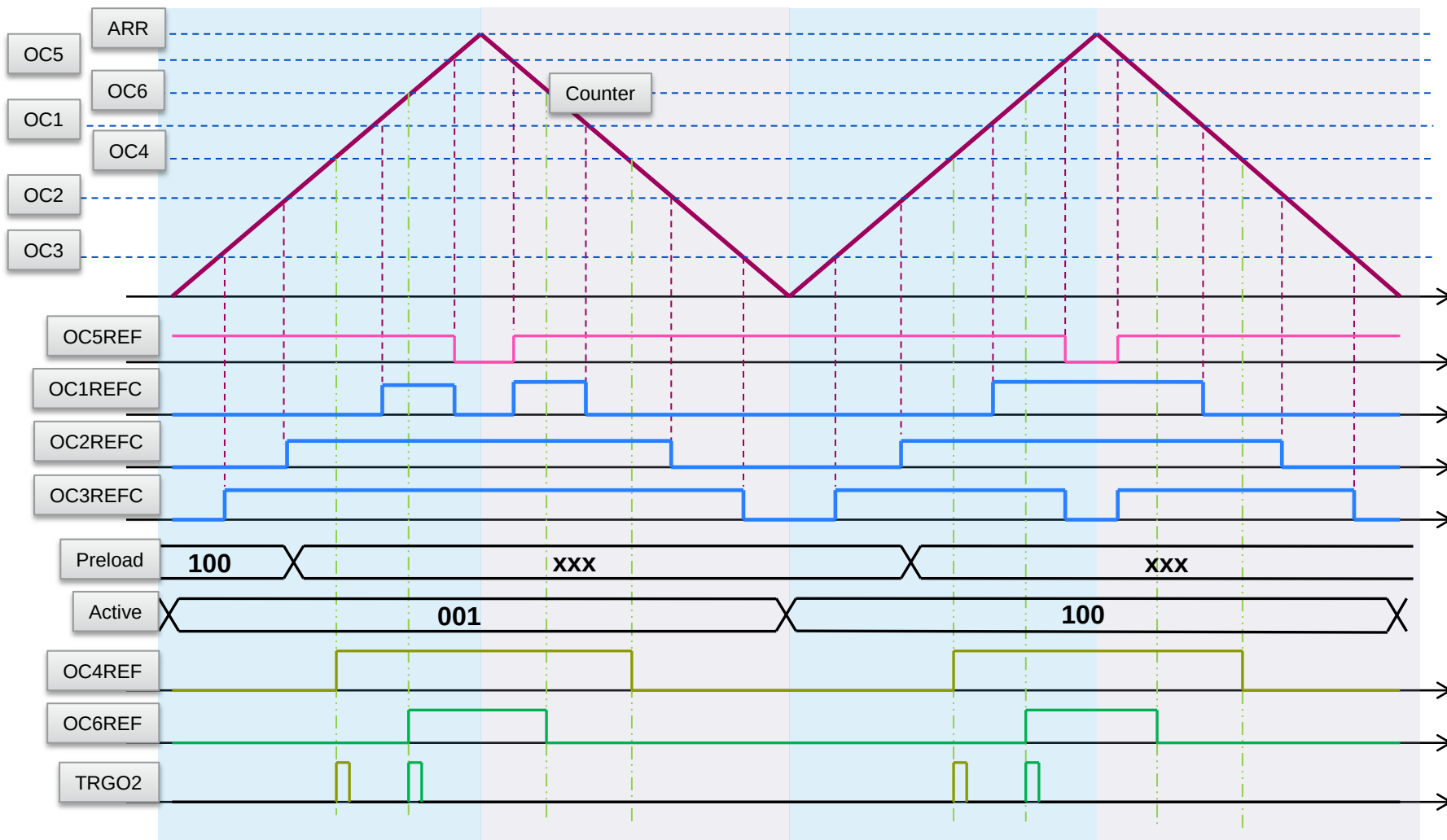
- Additional set of triggers dedicated for ADC
 - Outputted on the new (second) trigger output TRGO2
 - Controlled through the new bit-field MMS2[3:0]



Combined 3-phase PWM mode (1/3)

(Only TIM1 & TIM8)

19

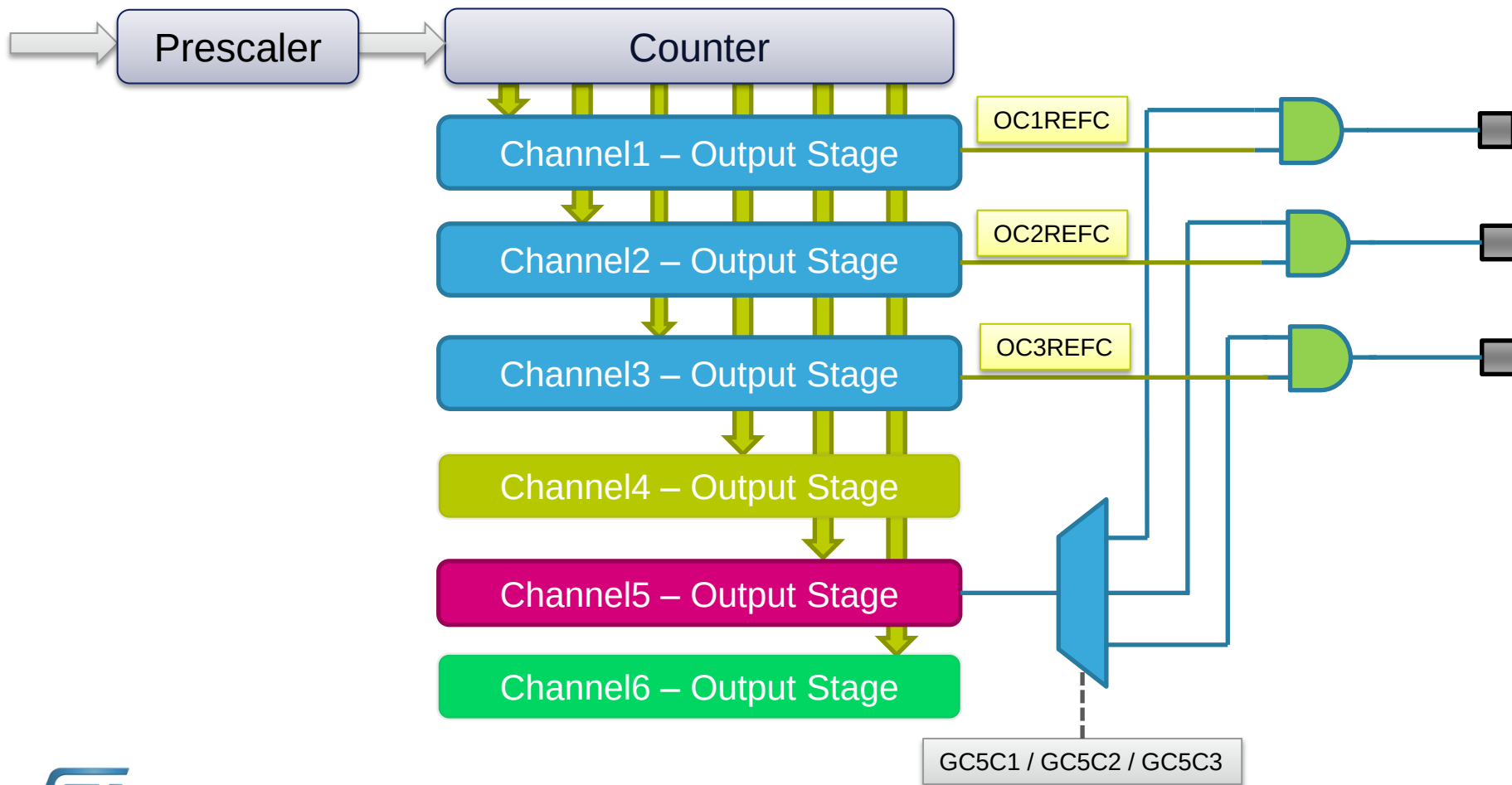


Combined 3-phase PWM mode (2/3)

(Only TIM1 & TIM8)

20

- Operation mechanism



Combined 3-phase PWM mode (3/3)

(Only TIM1 & TIM8)

21

- Waveforms generation on up to three channels
 - Based on coupling Channel5's output with others channels
 - Channel1
 - Channel2
 - Channel3
- Dedicated for Motor Control application
 - Used by ST's patented Single-shunt current reading application
 - Can reduce CPU load by 5-10% compared to current implementation on F1/F2/F4 families
 - Frees many MCU resources (DMA channels, Interrupt request lines)

Miscellaneous enhancements (1/2)

22

- Repetition counter width is up to 16 bit (Only TIM1/TIM8)
 - Gives about 650ms between updates for 100KHz PWM frequency
- Two OCxREF clearing sources:
 - External OCxREF clearing input: ETRF input
 - Internal OCxREF clearing input
 - Connected internally to comparator output: a pseudo-cell for Cycle-by-cycle current control
- Timers' synchronization enhancement
 - Introduction of a new synchronization mode: **Combined Reset+Trigger** Mode
 - When a trigger active edge is detected, the counter content is Reset and the counting is started
 - For configuring the *Retriggerable One-Pulse* mode, the timer has to be configured in slave mode: Combined Reset+Trigger mode shall be used

Miscellaneous enhancements (2/2)

23

- Up to two break input sources (Only TIM1/8)
 - Break input 1 (legacy one)
 - Idle State programming
 - Has the highest priority over Break inputs
 - Multiplexed with internal break signals:
 - Clock failure event from CSS block
 - SRAM parity error
 - Comparators outputs
 - PVD interrupt
 - Cortex M0 lockup (hard fault) output
 - Built with a digital filter with a flexible set of sampling periods
 - Asynchronous functioning (unless the filter is enabled)
 - Typical use case: Over-voltage protection handling
 - Break input 2 (new one)(only on TIM1/TIM8)
 - No Idle State programming
 - Lower priority compared to Break input 1 (legacy one)
 - Built with a digital filter with a flexible set of sampling periods
 - Typical use case: Over-current protection handling

Product-level enhancements (1/2)

24

- Two clock sources for Advanced Control Timers (TIM1/TIM8)
 - APB clock
 - PLL output
 - Advanced Control Timers operate with clock frequency up to 144MHz
 - To reach 144MHz operation frequency the following conditions shall be fulfilled:
 - SYSCK/AHB prescaler must be set to 1
 - AHB/APB prescaler must be set to 1
- Clock source selection
 - Please refer to TIMxSW (x = 1,8) bit description within RCC_CFGR3 register description (RCC chapter)
 - The TIMxSW (x = 1,8) control bit can set/reset by software
 - In case where one of the above conditions is not fulfilled, the TIMxSW control bit is reset by hardware

Product-level enhancements (2/2)

25

- Encoder Mode enhancement

- Two Timers can share the same Quadrature Encoder output signals
 - TIM2 IC1 (*respectively TIM2 IC2*) is connected to TIM15 IC1 (*respectively TIM15 IC2*)
 - TIM3 IC1 (*respectively TIM3 IC2*) is connected to TIM15 IC1 (*respectively TIM15 IC2*)
 - TIM4 IC1 (*respectively TIM4 IC2*) is connected to TIM15 IC1 (*respectively TIM15 IC2*)

- Configuration

- Using ENCODER_MODE bit field within the SYSCFG_CFGR1 register (for more details refer to SYSCFGR chapter)

- Use case

- Used with M/T technique for estimating Velocity and Acceleration for wide-range of velocity values (especially for low velocity values)



Thank you !

CONFIDENTIALITY OBLIGATIONS:

THIS DOCUMENT CONTAINS SENSITIVE INFORMATION.

IT IS CLASSIFIED "MICROCONTROLLERS, MEMORIES & SECURE MCUs (MMS) RESTRICTED AND ITS DISTRIBUTION IS SUBMITTED TO ST/MMS AUTHORIZATION

AT ALL TIMES YOU SHOULD COMPLY WITH THE FOLLOWING SECURITY RULES:

- DO NOT COPY OR REPRODUCE ALL OR PART OF THIS DOCUMENT
- KEEP THIS DOCUMENT LOCKED AWAY
- FURTHER COPIES CAN BE PROVIDED ON A "NEED TO KNOW BASIS", PLEASE CONTACT YOUR LOCAL ST SALES OFFICE OR DOCUMENT WRITTER.

Information furnished is believed to be accurate and reliable. However, STMicroelectronics assumes no responsibility for the consequences of use of such information nor for any infringement of patents or other rights of third parties which may result from its use. No license is granted by implication or otherwise under any patent or patent rights of STMicroelectronics. Specifications mentioned in this publication are subject to change without notice. This publication supersedes and replaces all information previously supplied. STMicroelectronics products are not authorized for use as critical components in life support devices or systems without express written approval of STMicroelectronics.

The ST logo is registered trademark of STMicroelectronics
All other names are the property of their respective owners

© 2015 STMicroelectronics - All Rights Reserved

STMicroelectronics group of companies Australia - Brazil - Canada - China - Finland - France - Germany - Hong Kong - India - Israel - Italy - Japan - Malaysia - Malta - Morocco - Singapore - Spain - Sweden - Switzerland - United Kingdom - United States.

www.st.com